Explainable AI-Based Anomaly Detection in Charging Station Networks: A Framework for EU AI Act Compliance

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Abstract. The rapid expansion of electric vehicle (EV) charging infrastructure poses substantial challenges for operational monitoring and fault diagnosis, yet existing methods rarely address cross-site generalization and regulatory compliance in a systematic manner. Based on 1.556 million sampling points (the full dataset) collected from 30 charging stations, this paper proposes an explainable anomaly detection framework. We introduce a multimodal attention fusion detector, Token-Attn, which integrates six-channel raw time series with 62-dimensional handcrafted features within a unified attention space, coupled with a three-view explainable attribution scheme (self-attention, permutation feature importance, and feature-group ablation). In cross-site evaluation on unseen sites (owners 7–8), Token-Attn attains PR-AUC = 0.918 and AUC = 0.990 via 7-seed probability ensembling, outperforming LightGBM (0.868, bootstrap p = 0.026) and exceeding XGBoost (0.887, p = 0.143). Attribution and fault subtyping further reveal that faults are essentially "abnormal termination under unattenuated high power" (electrical-interruption type 95.3% vs. over-temperature type 4.7%), with temperature rise being a consequence of charging duration rather than its cause. Building on these findings, we present a compliance framework aligned with the EU AI Act.

Keywords: explainable artificial intelligence; anomaly detection; EV charging stations; attention fusion; causal mechanism analysis; EU AI Act

1. Introduction

Deep learning has achieved strong performance in EV charging-station anomaly detection, yet two gaps remain. First, cross-site generalization: existing methods are mostly evaluated on distributionally consistent data, and a single global model degrades markedly on unseen sites [1]. Second, regulatory compliance: the EU AI Act [2] classifies AI systems used in critical infrastructure as high-risk, mandating transparency and explainability, whereas existing methods output only anomaly scores and cannot indicate which sensor triggered the alarm. We propose a multimodal attention fusion detector, Token-Attn, and validate its generalization under a cross-site protocol (owners 1–6 for training → owners 7–8 for testing), together with multi-view explainable attribution and causal mechanism analysis, ultimately providing a compliance scheme that maps obligations, audit logs, and model cards to the EU AI Act.

2. Method

Dataset. The dataset comprises 30 charging stations, 1.556 million sampling points (full dataset), and 8 data owners [1]. Charging sessions (≥ 30 sampling points) serve as classification units, yielding 19,658 valid sessions; owners 1–6 are used for training (13,505 sessions / 642 faults) and owners 7–8 form unseen test sites (2,776 sessions / 129 faults), simulating cold-start deployment.

Token-Attn. A bidirectional LSTM encodes the six-channel raw time series (voltage / current / power / gun temperatures 1/2 / SOC) and pools them into eight segment tokens by equally partitioning the charging progress; these tokens, together with 62-dimensional handcrafted feature tokens and a [CLS] token, are fed into a two-layer Transformer for cross-modal self-attention fusion, yielding binary classification (≈ 224K parameters).

Multi-view attribution. Self-attention + permutation feature importance (each feature column shuffled three times; the mean ± std of the PR-AUC drop is reported) + feature-group ablation, cross-validated across three views.

3. Experiments and Results

Cross-site performance on unseen test sites (owners 7–8) is reported in Table 1. With 7-seed probability ensembling, Token-Attn attains PR-AUC = 0.918 and AUC = 0.990, outperforming LightGBM (0.868, bootstrap p = 0.026) and exceeding XGBoost (0.887, p = 0.143), approximately 2.6× the end-to-end Bi-LSTM (0.351), indicating that attention-based interactive fusion is key to cross-site performance.

Table 1. Performance comparison on the cross-site test set (owners 7–8)

Method	Input	Recall	Precision	F1	AUC	PR-AUC
LSTM-AE (reconstruction,	6 ch	—	—	—	0.912	0.248

unsupervised)						
End-to-end Bi-LSTM	6 ch	77.52%	23.36%	0.359	0.908	0.351
DualTransformer	6 ch + 62-dim	25.58%	32.35%	0.286	0.758	0.242
iTransformer	6 ch + 62-dim	46.51%	66.67%	0.548	0.932	0.595
FT-Transformer	6 ch + 62-dim	57.36%	63.79%	0.604	0.950	0.624
PatchTST	6 ch + 62-dim	61.24%	71.17%	0.658	0.957	0.679
LightGBM	62-dim	31.78%	97.62%	0.480	0.987	0.868
XGBoost	62-dim	32.56%	95.45%	0.486	0.991	0.887
Token-Attn (7-seed ensemble)	**6 ch + 62-dim**	**68.99%**	**93.68%**	**0.795**	**0.990**	**0.918**

Note: all other deep models are single-seed (seed 42) results; the PR-AUC of every state-of-the-art deep model remains below that of GBDT. Token-Attn adopts a precision-prioritized default threshold, so its recall is slightly lower than that of Bi-LSTM while its precision and F1 are markedly better; for miss-sensitive scenarios the threshold can be lowered to increase recall.

Causal mechanism (derived from three-view attribution and 129 measured fault samples). Three-view attribution shows that gun-temperature statistics and basic statistics dominate discrimination, with attention drifting from early to mid-late charging segments under anomalies. The 129 test faults split into an electrical-interruption type (95.3%) and an over-temperature type (4.7%); faults are fundamentally “abnormal termination under unattenuated high power” (terminal SOC 84% vs. 98% normal), with temperature rise being a consequence of charging duration rather than its cause.

4. Conclusion

Contributions: (1) we reveal the fault fingerprint of “sudden interruption during the high-current high-power stage”; (2) we construct Token-Attn, which reaches PR-AUC = 0.918 cross-site, matching and partially exceeding GBDT; (3) we provide multi-view explainable attribution and causal mechanism analysis; and (4) we present an EU AI Act compliance framework (obligation mapping, audit logging, model cards [3]).

References

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